RBE 500 – Group Assignment – Part 1

OpenManipulator-X: Forward Kinematics, Inverse Kinematics, and Pick-and-Lift

110 / 100 points (10 points bonus)

In this assignment, your team will use an OpenManipulator-X robot arm to implement and test forward kinematics, inverse kinematics, and a simple pick-and-lift task using ROS 2.

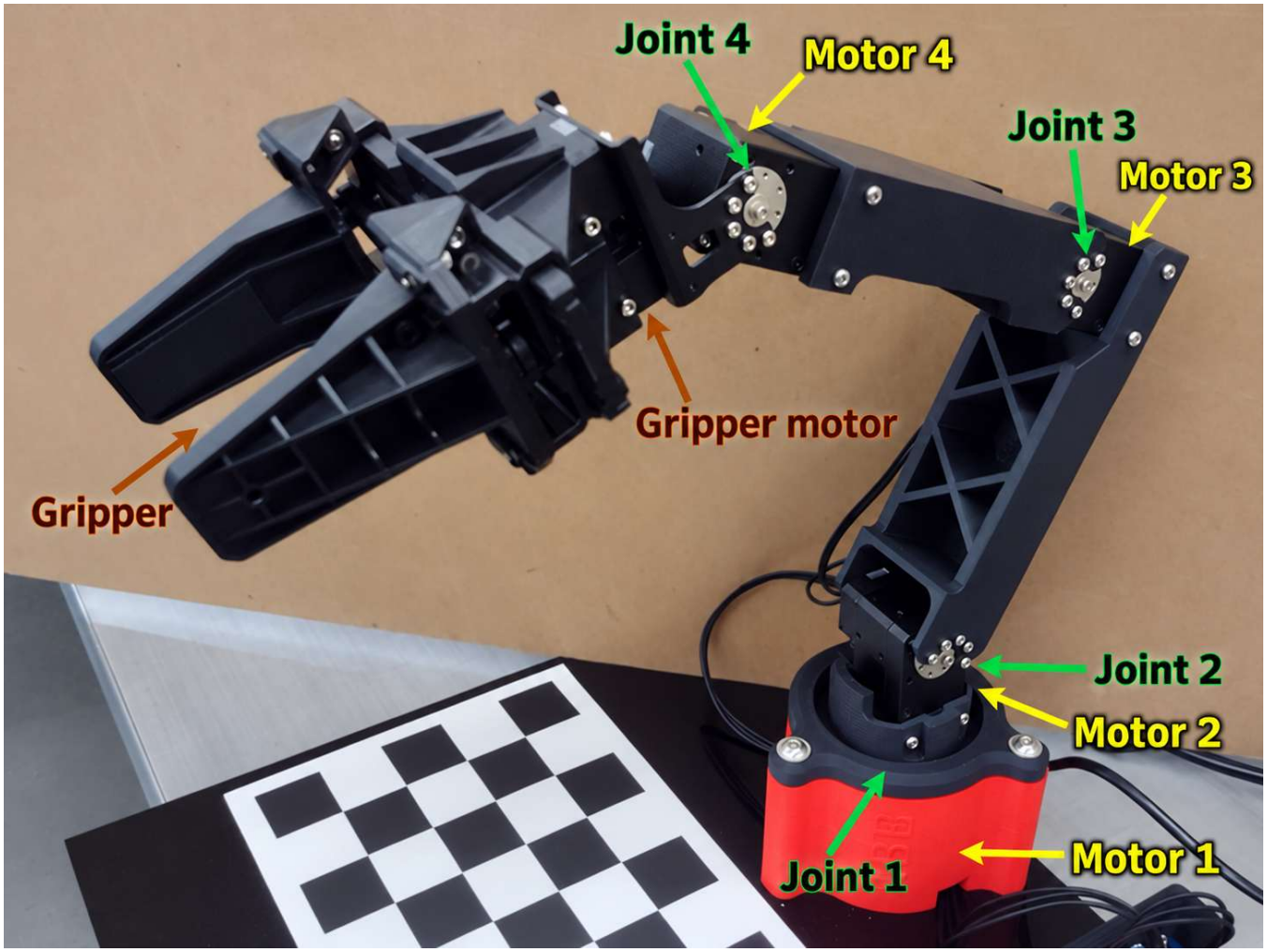


Figure 1. OpenManipulator-X robot arm used in this assignment.

General Instructions

- Each team will be assigned one robot arm and will use that same arm through the end of the semester.
- The robots are located in the RBE Lab in the Atwater Kent (AK) building room 120.
- The robot arms must remain in the RBE Lab at Atwater Kent. Do not remove them from the lab.
- You will have access to the AK120 lab to complete this assignment. The AK120 lab will be busy with the RBE 3002 students throughout the term. Labs are on Wednesday. You can work in AK120 if there's room, but if there's not, you can work elsewhere in AK like the Pumpkin Lounge or the tables in the basement. The arms just can't leave AK.
- Please return the arms to their shelves when done working.
- To help prevent breakages, keep motor speed low for initial testing to make sure code and positions work. Using torque control can cause issues if not careful. And be prepared to stop the arm in case it moves unexpectedly.

- If there are any issues with the arms, email gr-rbelas@wpi.edu and please do not forget to cc me.

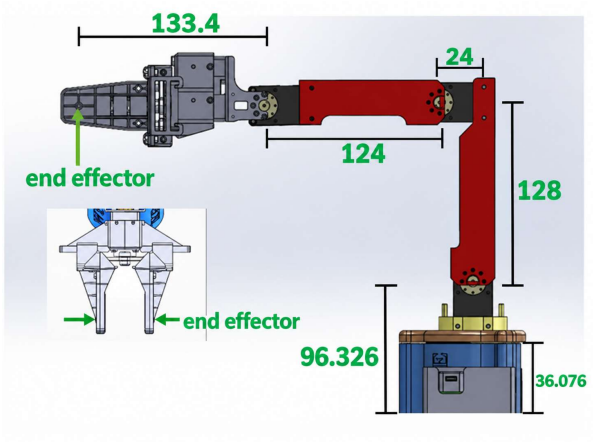
Important: Use consistent units throughout your kinematic calculations. The dimensions shown in the figures below are in millimeters.

1. Forward Kinematics (50 pts)

Implement a ROS 2 forward-kinematics node for the OpenManipulator-X arm. Your node should:

- Subscribe to the robot's joint-state topic and read the current joint positions.
- Calculate the end-effector pose with respect to the robot base frame using Denavit Hartenberg method. Here, you are **not** allowed to use high level libraries that will directly solve the forward kinematics for you. You can freely use math libraries.
- Publish the calculated end-effector pose on a ROS 2 topic. Please perform the calculation and publication inside the joint-state callback so that a new pose is published whenever new joint values are received.

Use the reference frames and dimensions shown below when developing your kinematic model.



Robot dimensions (all lengths in mm)

Testing

Move the robot to three distinct configurations by sending position references to the joints. For each configuration, use your forward-kinematics node to calculate the end-effector pose and display the published pose in the terminal using the `ros2 topic echo` command.

Deliverables

- Your ROS 2 package containing the forward-kinematics node.
- A brief description of your kinematic model and the transformations used to calculate the end-effector pose.
- For each of the three robot configurations: the commanded joint values, a screenshot/photo of the robot configuration, and a terminal screenshot showing the published end-effector pose using ros2 topic echo.

2. Inverse Kinematics (40 pts)

Implement inverse kinematics in a separate ROS 2 node using a service server. The service request should contain a desired end-effector pose, and the service response should contain the corresponding joint positions. You can use the terminal command `ros2 service call` as the service client when testing your implementation.

Testing

Test your inverse-kinematics service for three distinct, reachable end-effector target poses. For each test, record the requested pose and the joint positions returned by your service.

Deliverables

- Your ROS 2 package containing the inverse-kinematics service implementation, including any custom service definition that you create.
- For each of the three test poses: the desired end-effector pose and a terminal screenshot showing the ros2 service call request and the returned joint positions.
- A brief explanation of the inverse-kinematics method used in your implementation.

3. Pick and Lift (20 pts)

Place an object at a known location on the chessboard. Determine a suitable gripper pose for the object and use your inverse-kinematics implementation to calculate the required joint angles.

Use a pre-grasp pose located directly above the object. The robot should first move to the pre-grasp pose, then move down to the grasp pose, close the gripper, and return to the pre-grasp pose to lift the object.

Suggested motion sequence:

1. Move to the pre-grasp pose above the object.
2. Move vertically down to the grasp pose.
3. Close the gripper.
4. Move back to the pre-grasp pose while holding the object.

Deliverables

- In your report, state the known object location, the desired grasp pose, and the corresponding inverse-kinematics solution.
- A short video showing the complete pick-and-lift sequence.
- All code used to execute the pick-and-lift task.

Submission Checklist

Submit the following items for Part 1:

- A single PDF report containing the requested explanations, screenshots, robot configurations, test inputs, and results.
- Your complete ROS 2 package(s) and source code for the forward-kinematics, inverse-kinematics, and pick-and-lift implementations.
- The video of the pick-and-lift demonstration.